

# Michael Papamichael

London, UK • +44 7541483322 • m.papamichael29@gmail.com

[LinkedIn](#) • [GitHub](#)

Work Permit: UK Passport & EU Citizen

## Professional Summary

Quality assurance engineer and MSc graduate (Imperial College London) with 3+ years working at the intersection of financial data systems, statistical analysis, and software engineering. Experienced in Python, SQL, and applied machine learning, with hands-on delivery of predictive models, automated data pipelines, and large-scale financial dataset validation across FCA-regulated environments. Now focusing fully on data science and ML, bringing both the technical depth and financial domain expertise to deliver immediate impact.

## Work Experience

### Investment Software LTD | QA Engineer (Data, Automation & Technical Lead)

09/2022 – Present | London, UK

- Designed and executed complex SQL pipelines (CTEs, joins, query optimisation) to validate, map, and migrate AUM datasets scaling to billions of pounds across wealth management client onboardings — ensuring zero FCA CASS compliance failures prior to platform ingestion.
- Designed a 100K+ datapoint performance analytics framework that runs as an automated daily Power BI pipeline continuously monitoring system execution trends on bottlenecks and directly informing optimisations across releases.
- Built and maintained automated testing frameworks in C# with Selenium across UI and API layers, integrated into Azure DevOps CI/CD pipelines, ensuring data integrity and calculation consistency across development, UAT, and production.
- Technical lead on the company's new client-facing mobile application, holding full ownership of QA strategy, architecture, build, deployment, versioning decisions and security across the full development lifecycle.
- Led QA design and delivery of the company's back-office module from inception to client deployment, driving weekly stakeholder requirement sessions and ensuring scalability and robustness throughout.

### H2Go Power LTD | Data Scientist (Capstone Project during MSc)

06/2022 – 08/2022 | London, UK

- Developed an end-to-end ML pipeline utilising ARIMA time series forecasting to model raw material and logistics costs over a 10-year horizon, delivering the company's first quantitative supply chain planning framework to inform strategic investment decisions.
- Conducted comprehensive data acquisition and EDA, extracting market intelligence from multiple sources, performing feature engineering, and uncovering critical supply chain dependencies and cost drivers that underpinned the forecasting model.
- Delivered an executive Tableau dashboard communicating predictive findings, statistical findings and scenario analysis directly to the CTO, shaping strategic investment and supply chain decisions.

### Cyprus National Guard - Military Service | Master Sergeant

07/2017 – 09/2018 | Cyprus

- Advanced to supervisory rank via competitive assessment, managing a 20-person team across operations, shift planning, and training, improving readiness by 10%.

## Personal Projects

### UK Fuel Pricing Fairness Model | [Streamlit App](#) | [GitHub](#)

- Building an end-to-end overcharging detection model for UK petrol stations using the government's Fuel Finder open dataset: predicting the fair price each station should charge given wholesale costs, brand, and local competition, then flagging stations with large positive residuals.
- Engineered an automated data collection pipeline in Python using the Fuel Finder OAuth2 API, scheduling daily ingestion into date-partitioned Parquet queried with DuckDB SQL, and deployed a live Streamlit app early on to validate the pipeline.
- Engineering spatial competition features with GeoPandas, a LightGBM fair-price regression validated with spatial cross-validation and SHAP explanations, then extending the analysis with an asymmetric pass-through model in statsmodels.

### NY Real Estate Price Prediction | [GitHub](#)

- Built a ML pricing model using XGBoost and NLP feature engineering (TF-IDF, embeddings) on a messy dataset (77% missing values, KNN imputation) to predict NYC property prices from 8,000+ listings, achieving an  $R^2$  of 0.742.

## Education

### Imperial College Business School | MSc Business Analytics (Applied AI/ML) (Merit) | 2021 – 2022 | London, UK

Relevant Modules: Advanced Machine Learning, Financial Analytics, Statistics & Econometrics, Data Structures & Algorithms, Data Visualisation

### University of Manchester | BEng Mechanical Engineering (2:1) | 2018 – 2021 | Manchester, UK

Individual Project: Built an ML pipeline in Python integrating speech recognition models with Fortran-based engineering simulation software on Linux. The system used custom-trained speech recognition models to enable voice-controlled automation of simulation workflows, an early application of NLP before the current wave of AI tooling.

## Languages & Skills

**Languages:** English (Native), Greek (Native), French (Conversational)

**Programming & Tools:** Python, SQL, C#, Power BI (DAX, semantic modelling), Tableau, Azure DevOps, CI/CD

**ML & Data Science:** Regression, Classification, Time Series Forecasting, Clustering, XGBoost, NLP (TF-IDF, Sentence Embeddings), PCA, Regularisation, Feature Engineering, EDA

**Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Selenium